

Name : Adhithya.S
Reg. No: 192412352

CODE:

Mobile Price Prediction using Linear Regression

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Sample Mobile Dataset

```
data = {
    "RAM_GB": [
        2, 3, 4, 4, 6,
        6, 8, 8, 12, 12,
        16, 16
    ],
    "Storage_GB": [
        32, 32, 64, 128, 128,
        256, 128, 256, 256, 512,
        512, 1024
    ],
    "Battery_mAh": [
        3000, 3500, 4000, 4500, 4500,
        5000, 5000, 5000, 5500, 5500,
        6000, 6000
    ],
    "Camera_MP": [
        12, 16, 32, 48, 50,
        64, 50, 64, 108, 108,
        200, 200
    ],
    "Price": [
        8000, 10000, 14000, 18000, 22000,
        26000, 30000, 35000, 45000, 55000,
        70000, 90000
    ]
}
```

Convert dataset into DataFrame

```
df = pd.DataFrame(data)
```

```
print("MOBILE PRICE DATASET")
```

```
print("=" * 50)
```

```
print(df)
```

Input Features

```
X = df[
```

```

[
    "RAM_GB",
    "Storage_GB",
    "Battery_mAh",
    "Camera_MP"
]
]

# Target Variable
y = df["Price"]

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)

# Create Linear Regression Model
model = LinearRegression()

# Train the Model
model.fit(X_train, y_train)

# Predict Test Data
y_pred = model.predict(X_test)

# Calculate Performance
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# Display Results
print("\nMODEL PERFORMANCE")
print("=" * 50)

print("Mean Squared Error:", round(mse, 2))
print("R-squared Score:", round(r2, 4))

print("\nActual Prices:", list(y_test))
print(
    "Predicted Prices:",
    [round(price, 2) for price in y_pred]
)

# Predict Price for a New Mobile
new_mobile = pd.DataFrame({
    "RAM_GB": [8],
    "Storage_GB": [256],
    "Battery_mAh": [5000],
    "Camera_MP": [108]
})

predicted_price = model.predict(new_mobile)

```

```

print("\nNEW MOBILE DETAILS")
print("=" * 50)
print("RAM:", new_mobile["RAM_GB"].iloc[0], "GB")
print("Storage:", new_mobile["Storage_GB"].iloc[0], "GB")
print("Battery:", new_mobile["Battery_mAh"].iloc[0], "mAh")
print("Camera:", new_mobile["Camera_MP"].iloc[0], "MP")

print("\nPredicted Mobile Price: ₹",
      round(predicted_price[0], 2))

```

OUTPUT :

```

MOBILE PRICE DATASET
=====
   RAM_GB  Storage_GB  Battery_mAh  Camera_MP  Price
0        2         32         3000         12   8000
1        3         32         3500         16  10000
2        4         64         4000         32  14000
3        4        128         4500         48  18000
4        6        128         4500         50  22000
5        6        256         5000         64  26000
6        8        128         5000         50  30000
7        8        256         5000         64  35000
8       12        256         5500        108  45000
9       12        512         5500        108  55000
10      16        512         6000        200  70000
11      16       1024         6000        200  90000

MODEL PERFORMANCE
=====
Mean Squared Error: 8306082.2
R-squared Score: 0.9881

Actual Prices: [70000, 55000, 8000]
Predicted Prices: [np.float64(65678.83), np.float64(57391.59), np.float64(7274.69)]

NEW MOBILE DETAILS
=====
RAM: 8 GB
Storage: 256 GB
Battery: 5000 mAh
Camera: 108 MP

Predicted Mobile Price: ₹ 32182.76

```